

Ian M. Nesbitt

Williamstown, MA, USA

School of Earth and Climate Sciences, University of Maine

Member of the *Geodynamics Numerical Modeling Laboratory*

✉ ian.nesbitt@gmail.com

🌐 www.iannesbitt.org

🔊 [iannesbitt](#)

0000-0001-5828-6070

Last modified: May 13, 2026

Education

2017–2022 **Masters Student in Earth and Climate Sciences**, *School of Earth and Climate Sciences, College of Natural Sciences, Forestry, and Agriculture, University of Maine, Orono, ME, USA*

GPA: 3.97

Thesis: Using diverse methods to determine the volume and stratigraphy of sediment in a formerly glaciated lake in central Maine, USA

Advisor: [Seth W. Campbell](#), *Co-advisor:* [Sean M.C. Smith](#)

Associated Projects: [readgssi](#), a Python tool to read and process radar data; [SeidarT](#), a seismic and radar survey modeling toolbox

2009–2013 **Bachelor of Arts in Geosciences with Honors**, *Williams College, Williamstown, MA, USA*

Thesis: A comparative study of snowmelt-driven water budgets in adjacent alpine basins, Niwot Ridge, Colorado Front Range

Advisor: [David P. Dethier](#)

2006–2009 **Degree with Honors**, *Holderness School, Holderness, NH, USA*

Awards and Scholarships

2021 Golden Key (invited)

2020 US Antarctica Service Medal

2013 Sigma Xi induction

2012 Williams College Class of 1960 Scholar in Geosciences

Volunteer Experience

Since 2023 Nordic Ski Coach, Mt. Greylock Regional School

2022–2023 Assistant Nordic Ski Coach, Mt. Greylock Regional School

2018–2023 Co-President, Williams College Class of 2013

2017–2018 Co-Director of Reunion Planning, Williams College Class of 2013

Work Experience

2022–2026 **Research Software Engineer**, *National Center for Ecological Analysis and Synthesis (NCEAS), U.C. Santa Barbara, Santa Barbara, CA, USA (remote)*

Duties: Python and Java development of open-source cloud computing tools to interpret, quality-check, and archive large quantities of research science data.

Supervisor: [Matt Jones](#)

- 2022–2026 **Assistant Head Coach**, *Berkshire Nordic Ski Club*, Williamstown, MA, USA
Duties: Group and one-on-one summer Nordic ski training, technique instruction, goal setting, customized training plans. Planning and coordination with other coaches, athletes, and parents.
Head Coach: Hilary Greene
- 2022 **Geospatial Engineering Consultant**, *Muddy Water Initiative*, Boston, MA, USA
Duties: Geospatial database management, design of SQL joins and relates, QA/QC and digitization of field data, presentation-quality map design, client and stakeholder interaction for [Caroline Reeves](#).
- 2021 **Geophysicist**, *International Thwaites Glacier Collaboration, Project GHC*, Orono, ME, USA
Duties: Processed ground-penetrating radar (GPR) survey data from the Hudson Mountains, West Antarctica, in collaboration with [John Woodward](#), [Joanne S. Johnson](#), [Brent Goehring](#), and [Seth W. Campbell](#).
- 2021 **Research Software Developer**, *University of Maine*, Orono, ME, USA
Duties: Tested and debugged code for geophysical modeling software [SeidarT](#), developed a multi-platform install script for [Christopher C. Gerbi](#) and [Steven P. Bernsen](#).
Projects: <https://github.com/UMainedynamics/seidart>
- 2019–2020 **Lead Geophysicist**, *Allan Hills Antarctic Expedition I-165-M, Princeton University - NSF Award Number 1744993*, Allan Hills Field Camp, East Antarctica
Duties: Collected and processed 100 km of ground-penetrating radar (GPR) survey data from a blue ice area on the eastern side of the Trans-Antarctic Mountains. Deep field survival, data collection, processing, use of snow machines in harsh conditions.
- 2018–2020 **Lead Scientist**, *Raspberry Shake, S.A.*, Boquete, Chiriqui, Panama
Duties: Technical support, software development, QA/QC of seismic instruments, development and QA/QC of educational materials.
Projects: <https://github.com/raspishake/rsudp>, <https://edu.raspberrysshake.org>
- 2014–2017 **Geophysical Scientist**, *e4sciences LLC*, Newtown, CT, USA
Duties: Led field team for many types of geophysical survey including Mobile LiDAR, ground-penetrating radar, high-precision GPS, multibeam echosounding, sub-bottom seismic, sidescan sonar. Field interpretation, and on-the-ground operations decision-making including boat handling.
Projects: Created survey-to-delivery workflow for 3D LiDAR end-user deliverables.
- 2013–2014 **Assistant Nordic Ski Coach**, *St. Michael's College*, Colchester, VT, USA
Duties: Created and supervised individualized training plans for athletes, technique instruction, race day logistics including ski preparation.

Teaching

- 2019 **Ground-Penetrating Radar Theory and Applications to Geological Mapping – SC2**, *Northeastern Section Meeting, Geological Society of America*, Portland, ME, USA
 Seth W. Campbell and Ian M. Nesbitt (filling in for Steven A. Arcone) taught short course at NEGSA annual meeting, covering radar application, data collection, processing, and interpretation.
- 2018 **Global Environmental Change – ERS 201**, *University of Maine*, Orono, ME, USA
 Karl J. Kreutz and Ian M. Nesbitt instructed students in accepted and debated climate theory from an earth systems perspective.
- 2017 **Earth Systems – ERS 200**, *University of Maine*, Orono, ME, USA
 Aaron E. Putnam, Sean M.C. Smith, Peter O. Koons, and Ian M. Nesbitt instructed students in earth systems theory.
- 2012–2013 **GIS and Remote Sensing – GEOS 214**, *Williams College*, Williamstown, MA, USA
 David P. Dethier and Ian M. Nesbitt instructed students in traditional and emerging geomorphologic theory, methods, and practice.
- 2012 **Geomorphology – GEOS 201**, *Williams College*, Williamstown, MA, USA
 David P. Dethier and Ian M. Nesbitt instructed students in geographical information systems (GIS) theory, technology and software, and advised in the application of these methods via end-of-term projects.

Publications

Peer-reviewed scientific articles

- 2025 **Assessing the suitability of sites near Pine Island Glacier for subglacial bedrock drilling aimed at detecting Holocene retreat–readvance**, *The Cryosphere*
 Joanne S. Johnson, John Woodward, Ian M. Nesbitt, Kate Winter, Seth W. Campbell, Kier Nichols, Ryan Venturelli, Scott Braddock, Brent Goehring, Brenda L. Hall et al.
 doi: [10.5194/tc-19-303-2025](https://doi.org/10.5194/tc-19-303-2025)
- 2025 **Miocene and Pliocene ice and air from the Allan Hills blue ice area, East Antarctica**, *Proceedings of the National Academy of Sciences*
 Sarah Shackleton, Valens Hishamunda, L. Davidge, Ed Brook, J. Marks-Peterson, Austin Carter, S. Aarons, Andrei Kurbatov, Douglas Introne, Y. Yan, Ian M. Nesbitt et al.
 doi: [10.1073/pnas.2502681122](https://doi.org/10.1073/pnas.2502681122)
- in review **The sediment delivery continuum from deglaciation to the modern watershed based on lake sedimentary deposits in the Northeastern USA**, *Quaternary Research*
 Ian M. Nesbitt, Seth W. Campbell, Sean M.C. Smith, Bess G. Koffman, Steven A. Arcone, and Kristin M. Schild
 preprint: [10.13140/RG.2.2.11567.05281](https://doi.org/10.13140/RG.2.2.11567.05281)

- 2021 **rsudp: A Python package for real-time seismic monitoring with Raspberry Shake instruments**, *Journal of Open Source Software*
Ian M. Nesbitt, Richard I. Boaz, and Justin Long
doi: [10.21105/joss.02565](https://doi.org/10.21105/joss.02565)
- 2020 **Global quieting of high-frequency seismic noise due to COVID-19 pandemic lockdown measures**, *Science*
Thomas Lecocq, Stephen P Hicks, Koen Van Noten, Kasper van Wijk, Paula Koelemeijer, Raphael SM De Plaen, Frédéric Massin, Gregor Hillers, Robert E Anthony, Maria-Theresia Apoloner, ... Ian M. Nesbitt et al.
doi: [10.1126/science.abd2438](https://doi.org/10.1126/science.abd2438)
- 2016 **Tracing subarctic Pacific water masses with benthic foraminiferal stable isotopes during the LGM and late Pleistocene**, *Deep-Sea Research Part II: Topical Studies in Oceanography*
Mea S. Cook, A. Christina Ravelo, Alan C. Mix, Ian M. Nesbitt, and Nari V. Miller
doi: [10.1016/j.dsr2.2016.02.006](https://doi.org/10.1016/j.dsr2.2016.02.006)

Conference proceedings, presentations, and datasets

- 2022 **Post-glacial sediment delivery continuum to an impounded valley reach in central Maine: a multi-disciplinary approach**, *Geological Society of America Abstracts with Programs*, Invited talk
Ian M. Nesbitt, Sean M.C. Smith, Seth W. Campbell, Bess G. Koffman, Steven A. Arcone, and Kristin M. Schild
doi: [10.1130/abs/2022NE-374798](https://doi.org/10.1130/abs/2022NE-374798)
- 2021 **Unprecedented present deglaciation of Pine Island Glacier Inferred from Ice Penetrating Radar Studies of Localised Ice Domes in the Hudson Mountains**, *International Thwaites Glacier Collaboration Conference*, Poster
John Woodward, Joanne S. Johnson, Seth W. Campbell, Ian M. Nesbitt, Brenda L. Hall, Scott Braddock, Meghan Spoth, Brent Goehring, Ryan Venturelli, Dylan H. Rood, Kier Nichols, Johnathan Adams, and Greg Balco
- 2019 **A decision-making framework for sedimentation analyses in dammed river corridor impoundments**, *Geological Society of America Abstracts with Programs* 51, Portland, ME, Talk
Ian M. Nesbitt, Sean M.C. Smith, Bess G. Koffman, Seth W. Campbell, and Steven A. Arcone
doi: [10.1130/abs/2019NE-328587](https://doi.org/10.1130/abs/2019NE-328587)
- 2018 **Sedimentary architecture and accumulation rates of multiple lakes in New England, USA**, *AGU Fall Meeting Abstracts 2018*, Washington, D.C., Poster NS41B-0830
Ian M. Nesbitt, Seth W. Campbell, Steven A. Arcone, Sean M.C. Smith, and Bess G. Koffman
- 2018 **Sedimentary architecture of Maine lakes derived with ground-penetrating radar**, *Maine Water and Sustainability Conference*, Augusta, ME, Poster
Ian M. Nesbitt, Seth W. Campbell, Steven A. Arcone, and Sean M.C. Smith
- 2018 **Holocene sediment volume determined by ground-penetrating radar and sidescan sonar in Maine, USA**, *Geological Society of America, Northeastern Section - 53rd Annual Meeting*, Talk
Ian M. Nesbitt, Seth W. Campbell, Steven A. Arcone, and Sean M.C. Smith
doi: [10.1130/abs/2018NE-311370](https://doi.org/10.1130/abs/2018NE-311370)
- 2017 **Using ground-penetrating radar and sidescan sonar to compare lake bottom geology in New England**, *AGU Fall Meeting Abstracts 2017*, New Orleans, LA, Talk PP44B-01
Ian M. Nesbitt, Seth W. Campbell, Steven A. Arcone, and Sean M.C. Smith

- 2017 **New England lake bottom geology and sedimentation rates derived from ground-penetrating radar**, *GSA Annual Meeting*, Seattle, WA
Seth W. Campbell, Ian M. Nesbitt, Steven A. Arcone, Sean M.C. Smith
doi: [10.1130/abs/2017AM-306235](https://doi.org/10.1130/abs/2017AM-306235)
- 2015 **Geophysical imaging of dams and levees in the Northeast US**, *Society of Exploration Geophysicists Abstracts With Programs*, New Orleans, LA
Daniel A. Rosales, William F. Murphy IV, Matthew B. Art, Ian M. Nesbitt, Salvatore Triano, David C. Herron, Kurt Schollmeyer, James Trotta, W. Bruce Ward, Lisa Stewart, and William F. Murphy III
- 2014 **Tracing Bering Sea circulation with benthic foraminiferal stable isotopes during the Pleistocene**, *AGU Fall Meeting Abstracts 2014*, PP23D-08
Mea S. Cook, A. Christina Ravelo, Alan C. Mix, Ian M. Nesbitt, and Nari V. Miller
- 2013 **A comparative study of snowmelt-driven water budgets in adjacent alpine basins, Niwot Ridge, Colorado Front Range**, *Thesis, Department of Geosciences, Williams College*, Williamstown, MA
Ian M. Nesbitt
Advisor: David P. Dethier
Available via [Williams College Libraries](#)
doi: [10.13140/RG.2.2.33197.20966](https://doi.org/10.13140/RG.2.2.33197.20966)
- 2013 **A comparative study of snowmelt-driven water budgets in adjacent alpine basins, Niwot Ridge, Colorado Front Range**, *Geological Society of America, Northeastern Section - 48th Annual Meeting*, Bretton Woods, NH
Ian M. Nesbitt and David P. Dethier
- 2013 **A comparative study of snowmelt-driven water budgets in adjacent alpine basins, Niwot Ridge, Colorado Front Range**, *Twenty-sixth annual Keck Research Symposium in Geology, proceedings 2013*, Pomona, CA
Ian M. Nesbitt and Robert Varga

Open-source software

- 2021 **SeidarT: Seismic and radar survey modeling toolbox**
Steven P. Bernsen, Christopher C. Gerbi, Ian M. Nesbitt, Ann Hill, Senthil Vel, Knut Christianson, Seth W. Campbell, and Ben Hills
<https://github.com/UMainedynamics/SeidarT>
doi: [10.5281/zenodo.5498194](https://doi.org/10.5281/zenodo.5498194)
- 2020 **rsudp: Continuous visual display, sudden motion monitoring, and historical replay of Raspberry Shake data**
Ian M. Nesbitt, Richard I. Boaz, and Justin Long
<https://github.com/raspishake/rsudp>
doi: [10.21105/joss.02565](https://doi.org/10.21105/joss.02565)

- 2019 **gpx2dzg: a tool to convert GPX files to GSSI's proprietary DZG format, and plot comparisons of marks in GPX and DZT/DZX files**, *Zenodo*
 Developer and maintainer
<https://github.com/iannesbitt/gpx2dzg>
 doi: [10.5281/zenodo.3260948](https://doi.org/10.5281/zenodo.3260948)
- 2019 **readgssi: an open-source tool to read and plot GSSI ground-penetrating radar data**, *Zenodo*
 Developer and maintainer
<https://github.com/iannesbitt/readgssi>
 doi: [10.5281/zenodo.1439119](https://doi.org/10.5281/zenodo.1439119)

Technical Skills

Languages	Python, Matlab, Node.js, Ruby/Rails
Markup	Django Templates, HTML, CSS, $\text{\LaTeX}$, Markdown, Overleaf
Platform Tools	Docker, Heroku, AWS
Developer Tools	GNU/Linux, Unix Terminal/bash/zsh, GNU Make, git, GitHub, Jupyter Notebooks, Visual Studio Code, Apple Developer Tools
Bug Tracking	MantisBT, GitHub
APIs	FDSN , USGS Earthquakes/EarthExplorer, Twitter, Telegram, Discord, Google Maps, Google Workspace
Other Tools	QGIS, ArcGIS, Inkscape, GIMP, Illustrator, Photoshop, ReflexW, RADAN
Communication	Slack, Discord, Mattermost
Office Suites	Microsoft Office, LibreOffice, Google Docs/Sheets/Slides

Spoken Languages

English	Native
Spanish	Beginner

Certifications

2013, 2018	Wilderness First-Aid
2016	CPR, First-Aid, and Defibrillator
2019	Antarctic Deep Field Survival
2020	T3 Alliance certified instructor